

Name \_\_\_\_\_ Date \_\_\_\_\_

# 9.1 Databases

## Getting started

Complete the table for the following single-table database.

| ID | Date      | Time  | Order Number |
|----|-----------|-------|--------------|
| 15 | 9/9/2020  | 15:00 | 2A           |
| 16 | 9/9/2020  | 15:05 | 2B           |
| 17 | 9/9/2020  | 15:11 | 2C           |
| 18 | 9/9/2020  | 15:15 | 2D           |
| 19 | 10/9/2020 | 10:00 | 3A           |
| 20 | 10/9/2020 | 10:30 | 3B           |

|                               |  |
|-------------------------------|--|
| Number of fields              |  |
| Number of records             |  |
| The primary key               |  |
| Number of orders on 9/9/2020  |  |
| Number of orders before 12:00 |  |
| Data type for ID              |  |
| Data type for Time            |  |
| Data type for Order Number    |  |

## Practice

A job centre stores details about the jobs that are currently on offer. This information includes: a unique job number, the company the job is with, whether it is part or full time, the yearly salary.

Define a single-table database for the job centre by identifying the fields, data types, and primary key.

## Challenge

Some databases need more than one table. Databases can also be set up to link tables together, so if you look at the data in one table it will align with the data in the other table. This is called a relationship.

Find an example where a database has more than one table, and where the tables are joined with a relationship. Identify the key components of the database tables, and how the relationship is set up between them.

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## 9.2 SQL

These questions use the following database table, APPOINTMENT.

| Appointment ID | CustomerID | Date       | Time  | Attended? | Fee    |
|----------------|------------|------------|-------|-----------|--------|
| 56             | 3DF        | 10/10/2020 | 09:30 | Yes       | 125.00 |
| 57             | 6HF        | 10/10/2020 | 10:45 | Yes       | 220.00 |
| 58             | 7VD        | 11/10/2020 | 10:00 | No        | 100.00 |
| 59             | 2SD        | 11/10/2020 | 12:15 | No        | 160.00 |
| 60             | 9JK        | 11/10/2020 | 15:00 | Yes       | 200.00 |

### Getting started

Match the SQL queries to their output.

```
SELECT CustomerID
FROM APPOINTMENT
```

3DF  
6HF  
9JK

```
SELECT AppointmentID
FROM APPOINTMENT
ORDER BY Time
```

3DF  
6HF  
7VD  
2SD  
9JK

```
SELECT CustomerID
FROM APPOINTMENT
WHERE Attended? = Yes
```

56  
58  
57  
59  
60

```
SELECT AppointmentID
FROM APPOINTMENT
WHERE Date = 10/10/2020
ORDER BY Time Desc
```

57  
56

Match the following SQL queries to their output.

```
SELECT COUNT(Time)
FROM APPOINTMENT
```

260

```
SELECT SUM(Fee)
FROM APPOINTMENT
WHERE Attended? = No
```

2

```
SELECT SUM(Fee)
FROM APPOINTMENT
WHERE Date = 10/10/2020
```

5

```
SELECT COUNT(FEE)
FROM APPOINTMENT
WHERE FEE >= 200
```

345.00

## Practice

Identify the output from the following SQL scripts:

- 1 

```
SELECT Appointment ID, CustomerID
FROM APPOINTMENT
WHERE Date = 11/10/2020 AND Attended = No
```
- 2 

```
SELECT Date, Time
FROM APPOINTMENT
WHERE Fee < 200
ORDER BY Fee DESC
```

- 3** SELECT Appointment ID, CustomerID, Time, Attended?  
FROM APPOINTMENT  
WHERE Fee > 100 AND Fee < 200
- 4** SELECT Appointment ID, CustomerID  
FROM APPOINTMENT  
WHERE Date = 10/10/2020  
ORDER BY Fee
- 5** SELECT COUNT(Appointment ID)  
FROM APPOINTMENT  
WHERE Attended? = No
- 6** SELECT SUM(Fee)  
FROM APPOINTMENT  
WHERE Time < 12:00

## Challenge

Complete the following SQL scripts.

- 1** Return the Customer IDs of all customers attended an appointment on the 10/10/2020.

```
_____ CustomerID  
_____  
_____  
_____ Date = 10/10/2020
```

- 2** Return the Appointment IDs of customer 7VD

```
SELECT _____  
FROM _____  
WHERE Customer _____ "7VD"
```

- 3** Return the Date and Time of all appointments on the 11/10/2020 where the fee is less than 100.00

```
SELECT _____ , _____  
FROM APPOINTMENT  
WHERE Date _____ 11/10/2020  
_____ Fee _____ 100
```

- 4** Return the Fee of all customers who attended appointments on the 10/10/2020 in descending order of Time.

SELECT Fee

FROM APPOINTMENT

WHERE \_\_\_\_\_

\_\_\_\_\_ Time \_\_\_\_\_

- 5** Return the total cost of all appointments.

SELECT \_\_\_\_\_ (\_\_\_\_\_)

FROM APPOINTMENT

- 6** Return the number of appointments between 10:00 and 13:00 inclusive.

SELECT \_\_\_\_\_ (AppointmentID)

FROM APPOINTMENT

WHERE Time \_\_\_\_\_ 10:00 \_\_\_\_\_ Time \_\_\_\_\_ 13:00

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## 9.3 Database

A theatre has a number of shows booked for upcoming months. Only one show takes place on each date. The theatre stores information about the shows in a database including:

- date
- show name
- type, e.g. comedy
- length.

It also stores the tickets that have been booked in a database including:

- booking ID
- date
- number of tickets
- name for booking
- telephone number
- paid (Yes or No).

### Getting started

Identify an appropriate field name for each of the data items in each of the two database tables.

Give each table a suitable name.

Select an appropriate data type for each of the fields.

Select an appropriate primary key for the database.

### Practice

Create two database tables in the database software of your choice, using the information you decided in the Getting started activity.

Write SQL scripts to return:

- 1 the number of tickets booked for a specific date
- 2 the telephone numbers of all bookings that have not been paid for
- 3 all of the shows in December.

## Challenge

Identify **four** pieces of information you want to gather from the data.  
Write an SQL script for each of these requirements.